Predictive Information in Conjunction Data-Message Histories Decays as Time to Closest Approach Decreases

Arjun Vijay Prakash
City Montessori School, Kanpur Road Campus
Lucknow, India

***Abstract*—Conjunction Data Messages (CDMs) report a collision probability that moves as later tracking shrinks orbital uncertainty. Operators often start planning about two days before closest approach, so an earlier forecast is useful only if the history available then contains information beyond copying the latest report. Using the public ESA Collision Avoidance Challenge archive, this study freezes the clock at 72, 48, 24, and 12 hours before closest approach and asks whether cutoff-safe summaries improve a forecast of the later reported log10 collision probability over persistence. On a leakage-safe hold-out of 1659 events, 1352 later reports sit at the dataset floor of -30. Unguarded XGBoost lowers overall mean absolute error from 5.080 to 2.809 (95% bootstrap interval on the difference $[1.930, 2.638]$). Across five grouped event-split redraws the MAE advantage is 2.34 ± 0.04 (seed 42 remains 2.27). Floor-excluded MAE rises from 4.073 to 7.562, persistence median AE is already 0, and a paired Wilcoxon test on per-event absolute error does not reject equality ($p = 0.91$). The selected ensemble with a -6 persistence guard has overall MAE 3.059 ($[1.716, 2.380]$) and Wilcoxon $p = 0.31$. Predicting $y - \mathrm{risk}$ and reconstructing $\mathrm{risk} + \widehat{\Delta}$ reaches test MAE 2.760. A two-part floor model (classifier for $P(y = -30)$, residual regressor on non-floor training events, threshold 0.15 chosen on validation) reaches test MAE 2.109 ($[2.524, 3.415]$) and median AE 0, with Wilcoxon $p < 10^{-33}$ versus persistence, but floor-excluded MAE rises to 9.311 and F_2 remains 0. It is a stored candidate, not the live exhibit. ESA-style loss (high-risk mean squared error divided by F2) ties at 0.167 for persistence and the selected ensemble. Leave-one-out on all 66 local high-risk events: persistence is closer than residual reconstruction on $66/66$ (mean absolute error 1.21 versus 12.59). The learned MAE advantage is largest at 72 hours and nearly gone at 12 hours. Bootstrap 90 percent bands cover 47.7 percent of outcomes and are therefore model spread. Split-conformal 90 percent intervals, fit on 1244 calibration events, cover 89.7 percent of the untouched test outcomes (mean width 18.33 log units). On ESA's 2167 official-test events (150 high-risk; labels released 2021, scored once after freeze), clipped persistence matches Uriot's last-risk-prediction loss $L = 0.694$. The selected ensemble ties that loss via the persistence guard; residual and floor candidates lower MAE but raise L to 104 and 70. Combined covariance volume at T-48 correlates with later $|\Delta\mathrm{risk}|$ ($\rho = 0.40$); the max-risk gap correlates with floor labels ($\rho = 0.41$) but with *less* movement ($\rho = -0.77$). The same frozen forecasts, rescored only by class cut, still tie persistence at -5 ($n+ = 3$); at -4 ($n+ = 1$) every system has $F_2 = 0$. This is a research prototype, not flight software.**

***Index Terms*—conjunction data message, collision probability, space debris, forecast horizon, gradient boosting, persistence**

Living manuscript (v2). Results are the 18 August 2026 freeze until later phases replace them. Unresolved items are marked as not yet measured.

I. Introduction

Two orbiting objects can pass close enough to generate a conjunction alert. The reported collision probability P_c is not a physical constant: it is recomputed as radar and optical observations shrink (or fail to shrink) the combined covariance [1], [2]. Waiting usually improves the estimate and leaves less time to plan. ESA posed the forecasting form of this problem in the 2019 Collision Avoidance Challenge: predict the final reported $\log_{10} P_c$ from messages with time to closest approach of at least two days [3], [4]. Persistence—copying the latest pre-cutoff risk—was a strong baseline. Only 12 of 96 teams beat it on ESA-style loss.

That competition does not answer how the extra information, if any, depends on remaining time, nor whether a mean-error win is a typical-event win. Physics-based P_c forecasting via expected covariance reduction [5] and operational work on probability dilution [2], [6] suggest that large early covariances can make P_c look safely small, then later updates either collapse risk to negligible or concentrate it. A recent machine-learning model predicts which conjunctions will keep high covariance, for extra tracking [7]. The target here is different: the later reported $\log_{10} P_c$ itself, under a frozen information cutoff.

The working hypotheses are as follows.

- H1: Extra predictive information beyond copying the latest report is largest at long horizons and approaches zero near closest approach.
- H2: Most of the mean-error win is rare floor jumps, not typical events becoming more accurate.
- H3: Extra information does not automatically improve high-risk decisions under the ESA class $\log_{10} P_c \geq -6$.
- H4: Large combined covariance at T-48 predicts later $|\Delta\mathrm{risk}|$. The snapshot gap `max_risk_estimate - risk` predicts floor membership, not larger later movement.

II. Methods

A. Data

Labeled CDMs from the ESA training archive comprise 162,634 rows and 13,154 events, 2015–2019, anonymized [4]. Events need a message at or before the cutoff and a later labeled update. That yields 8,293 eligible events at 48 hours before closest approach. Train, validation, calibration, and test are event-disjoint (3,731 / 1,659 / 1,244 / 1,659). Seed 42 is the reported local split. Five grouped redraws use seeds 42–46 with the same recipe and no test-tuned knobs; those rows are extra measurements, not extra models. Leave-one-high-risk-out retrains residual XGBoost on all eligible events except one of the 66 local $\log_{10} P_c \geq -6$ events, then scores that event. Features use only rows with time to closest approach at or above the cutoff. The later $\log_{10} P_c$ is the label, never an input. Official-test inputs are the public Kelvins `test_data.csv` (all $\mathrm{time\_to\_tca} \geq 2$). Labels are the released `true_risk` column from Zenodo record 4463683 (25 January 2021), joined after freeze. Shared numeric `event_id`s between train and test are independent numbering, not shared conjunctions (0 identical pre-cutoff snapshots). Private post-cutoff fields are not features. No hyperparameter was changed after seeing official-test scores.

B. Model

The current selected policy on the project surface is event-level XGBoost on inspectable snapshot and history summaries, with a ten-model bootstrap median and a persistence guard when the current report is already at or above -6. Baselines are persistence, the training-set median, and Ridge. A residual candidate predicts $y - \mathrm{risk}$ on cutoff-safe training rows and reconstructs $\widehat{y} = \mathrm{risk} + \widehat{\Delta}$. A two-part floor candidate trains a classifier for $P(y = -30)$ from the final reported risk, fits the residual regressor on non-floor training events only, and at inference predicts -30 when $P(\mathrm{floor})$ exceeds a threshold chosen on validation (here 0.15); a -6 persistence guard was offered as a component and rejected on validation MAE. Both candidates are stored artifacts and are not the live exhibit. Official-test scoring uses those frozen artifacts only. Predictions below -6 are clipped to -6.001 for ESA-style loss, matching Uriot *et al.* [3]. A class-threshold sweep reuses those frozen point forecasts and changes only the class cut at $\log_{10} P_c \in \{-8, -7, -6, -5, -4\}$; models are not retuned. For each cut, predictions below t are clipped to $t - 0.001$.

C. Metrics

Reported scores are mean and median absolute error in $\log_{10} P_c$, the share within 0.5 log units, and ESA-style loss $L = \mathrm{MSE}_{HR}/F_2$ with $\beta = 2$ on the -6 class [3]. Operational LEO reaction is nearer -4 to -5; ESA scored -6 to keep enough positives. The sweep reports L, F_2, $n+$, and a false-reassurance analogue (accepted forecast under the existing -6 abstention mask with $\widehat{y} < t$ while $y \geq t$). L is omitted when $F_2 = 0$. An event is a floor event if the later report satisfies $y \leq -30+10^{-6}$. Floor-excluded MAE is MAE on the complement. Residual MAE is $\mathrm{mean}|y - \mathrm{risk}|$ versus $\mathrm{mean}|\widehat{y} - \mathrm{risk}|$: how far the later report moved, versus how far the model moved the snapshot. A 95% interval on MAE(persistence)$-$MAE(model) uses 1000 event-resamples. A two-sided Wilcoxon signed-rank test compares per-event absolute errors. Split conformal uses absolute residuals $|y - \widehat{y}|$ of the exhibit ensemble on the 1244 calibration events [8]. Official-test L is the secondary evaluation on 2167 events with 150 high-risk positives. The dilution probe is a logistic / Spearman baseline, not an extra XGBoost model: `dilution_gap` $=$ `max_risk_estimate` $-$ `risk` is excluded from the exhibit feature list. Spearman correlations are reported on the frozen test. A logistic model of floor collapse on `dilution_gap`, miss distance, and message count is fit on train only; AUC is on test. Quartile edges for the gap table come from train. This probe follows Alfano and CARA dilution [2], [6]; it does not predict residual covariance for extra tracking [7].

Honest metrics, residual and floor candidates, split-conformal coverage, and the class-threshold sweep were scored on the frozen 18 August split. The homepage still shows one bootstrap band, labeled as model spread. It does not offer a threshold picker. The laboratory page reports conformal coverage; it does not add a calibration chart to the exhibit queue.

III. Results

Table I reports the frozen local test with floor slices and intervals. Of 1659 events, 1352 later reports sit at -30. Persistence median AE is 0: more than half of reports do not move after 48 hours. Reconstructing $\widehat{y} = \mathrm{risk} + \widehat{y - \mathrm{risk}}$ tests whether anything besides persistence exists. Residual XGBoost reaches test MAE 2.760 ($\Delta = 2.319$ $[1.977, 2.685]$), close to unguarded level-valued XGBoost (2.809). The two-part floor model reaches test MAE 2.109 ($\Delta = 2.971$ $[2.524, 3.415]$) and median AE 0. Floor-class confusion at the validation threshold 0.15 is 1319 true positives, 174 false positives, 33 false negatives, and 133 true negatives (recall 0.976, precision 0.883). Floor-only MAE falls to 0.474, but floor-excluded MAE rises to 9.311 versus persistence 4.073: the mean and Wilcoxon wins ($p < 10^{-33}$) come from calling the floor, including on 174 non-floor events. On validation, this candidate ranks first by MAE (1.916 versus unguarded XGBoost 2.669) and remains a stored candidate, not the live exhibit. $F_2 = 0$ for the floor model, residual model, and unguarded XGBoost. ESA-style loss still ties at 0.167 for persistence and the selected ensemble (H3). Across five grouped redraws, unguarded XGBoost MAE advantage is 2.34 ± 0.04 and floor-excluded MAE is 7.78 ± 0.21 (persistence floor-excluded 4.32 ± 0.23). Residual reconstruction is 2.35 ± 0.04 MAE advantage. Seed 42 remains the reported row in Table I. Leave-one-out on the 66 high-risk eligible events: persistence is closer than residual on $66/66$ (mean absolute error 1.21 versus 12.59). The constant training-set median still beats the selected policy on overall MAE (3.002).

Table II and Fig. 1 support H1 on this freeze: the learned advantage shrinks as closest approach approaches. The 12-hour gap is 0.06 log units and is *not yet* equipped with a confidence interval. The T$-$48 MAE gap has a within-split bootstrap interval (Table I) and a 2.34 ± 0.04 range across five grouped redraws.

Snapshot-only MAE is 2.904; adding historical summaries of the same fields lowers it to 2.851; covariance trends reach

TABLE I

FROZEN LOCAL TEST AT 48 HOURS ($n = 1659$; 1352 FLOOR EVENTS). MAE IN $\log_{10} P_c$. Δ IS MAE(PERSISTENCE)−MAE(MODEL) WITH A 95% BOOTSTRAP INTERVAL (1000 EVENT RESAMPLES). WILCOXON p IS TWO-SIDED ON PAIRED ABSOLUTE ERROR. SELECTED POLICY: BOOTSTRAP ENSEMBLE WITH A -6 PERSISTENCE GUARD.

System	**MAE**	**Med.**	**Non-fl.**	**Floor**	**Δ [95%]**	p
Persistence	5.080	0.000	4.073	5.308	—	—
Unguarded XGB	2.809	0.554	7.562	1.730	2.270 [1.930, 2.638]	0.91
Residual XGB	2.760	0.453	7.375	1.712	2.319 [1.977, 2.685]	0.92
Floor hurdle	2.109	0.000	9.311	0.474	2.971 [2.524, 3.415]	$< 10^{-33}$
Selected ens.	3.059	0.473	7.332	2.088	2.021 [1.716, 2.380]	0.31

TABLE II

SINGLE XGBOOST VERSUS PERSISTENCE BY HORIZON. MAE IN $\log_{10} P_c$. THE 72 H AND 12 H ELIGIBLE SETS OVERLAP THE 48 H SPLIT BUT ARE NOT IDENTICAL.

Horizon	**XGBoost**	**Persist.**	**Adv.**
72 h	3.214	7.748	4.534
48 h	2.808	5.080	2.272
24 h	2.110	2.634	0.524
12 h	1.384	1.444	0.060

2.808. History therefore adds a small increment that also still needs an interval.

Nominal 50% and 90% bootstrap bands cover 26.0% and 47.7% of outcomes (mean widths 0.90 and 2.05). Split-conformal bands around the same exhibit point cover 49.6% and 89.7% (mean widths 0.93 and 18.33). Table III and Fig. 2 record that comparison. The 90% conformal radius is 9.17 log units because floor jumps of twenty units sit in the calibration scores. Crossing -6 with the conformal 90% band was offered as an abstention rule and rejected on validation false reassurance; the live exhibit keeps the bootstrap-spread rule (test false reassurance $= 1$ of 9, versus 2 for the conformal alternative). One accepted high-risk miss remains. A four-mission hold-out has high-risk MAE 19.2 on a single held-out high-risk event.

Table IV and Fig. 3 report the one-shot official-test score. Persistence $L = 0.694$ matches the last-risk-prediction number in Uriot *et al.* on this distribution. The selected ensemble ties that loss because the -6 guard copies the snapshot whenever it is already high-risk. Residual and floor candidates still reduce MAE (3.476 and 3.177 versus 5.209) and the floor hurdle Wilcoxon versus persistence is $p < 10^{-17}$, but F_2 collapses and L rises to 104 and 70. Unguarded XGBoost has $F_2 = 0$. None of these frozen models beats the published sesc $L = 0.556$. That is the result; the models were not retuned.

Table V and Fig. 4 report H4. Combined $\log\det$ of the snapshot covariance correlates with later $|y - \text{risk}|$ ($\rho = 0.399$, $p < 10^{-63}$). F_{10} is weaker ($\rho = 0.108$). The max-risk gap is strongly associated with ending at the floor ($\rho = 0.407$; logistic test AUC 0.819) but with *less* later movement ($\rho = -0.768$): large gaps are mostly snapshots already near -30. Quartile 1 (small gap) has mean $|y - \text{risk}| = 13.03$ and floor rate 0.56; quartiles 3–4 have floor rates above 0.95 and

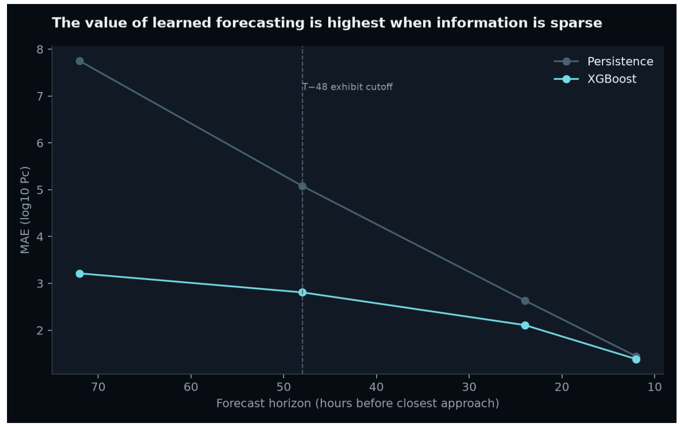

Fig. 1. Forecast-horizon MAE for the learned model and persistence (18 August freeze).

TABLE III

INTERVAL COVERAGE ON THE FROZEN LOCAL TEST ($n = 1659$). BOOTSTRAP BANDS ARE MODEL SPREAD. SPLIT CONFORMAL IS FIT ON 1244 CALIBRATION EVENTS ONLY.

Band	**Nom.**	**Cover.**	**Width**
Bootstrap spread	50%	26.0%	0.90
Bootstrap spread	90%	47.7%	2.05
Split conformal	50%	49.6%	0.93
Split conformal	90%	89.7%	18.33

mean movement below 0.75. The naive story that a large `max_risk` − `risk` gap means the report will still move is rejected on this freeze. Large covariance volume predicting more movement is supported.

Table VI reuses the frozen local-test forecasts and changes only the class definition. Operational LEO reaction is nearer -4 to -5; ESA scored -6 so the positive class was large enough to estimate F_2. Persistence and the selected ensemble still tie at -6 ($n+ = 9$, $L = 0.167$, $F_2 = 0.361$, one accepted false reassurance) and at -5 ($n+ = 3$, $L = 0.032$, $F_2 = 0.349$, zero accepted false reassurances). At -8 and -7 persistence has higher F_2 and lower L than the ensemble. At -4 a single test event is positive and every frozen system misses it ($F_2 = 0$). Unguarded XGBoost, residual reconstruction, and the floor hurdle have $F_2 = 0$ at -7 through -4.

IV. DISCUSSION

The 2019 challenge already showed that beating persistence on ESA-style loss is hard [3]. The contribution attempted here is not a second leaderboard entry. It is a measurement of *when* CDM history still contains extra information, and of the fact that mean error and high-risk decisions are different jobs. Large early covariance can dilute P_c [6]; later contraction often drops reported risk [2]. That physical picture matches a horizon-dependent learned advantage and a mean wrecked by floor collapses: 1352 of 1659 test labels are at -30, floor-excluded MAE is worse than persistence, and Wilcoxon tests on paired absolute error do not reject equality. It does not match a claim of operational warning skill on nine local

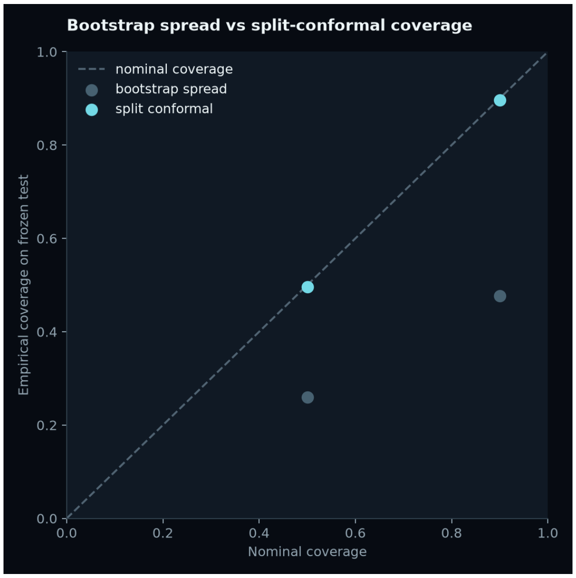

Fig. 2. Nominal versus empirical coverage for bootstrap spread and split conformal (frozen test).

TABLE IV
OFFICIAL TEST ($n = 2167$; 150 HIGH-RISK; 1673 FLOOR). MAE IN $\log_{10} P_c$. $L = \mathrm{MSE}_{HR}/F_2$ AFTER CLIPPING PREDICTIONS BELOW -6 TO -6.001. FROZEN BEFORE LOOK.

System	MAE	Med.	Non-fl.	L	F_2
Persistence (LRP)	5.209	0.000	4.287	0.694	0.739
Unguarded XGB	3.504	0.690	9.333	1.75×10^6	0.000
Residual XGB	3.476	0.594	9.289	104	0.017
Floor hurdle	3.177	0.000	11.832	70.5	0.025
Selected ens.	3.107	0.439	6.750	0.694	0.739

positives or on leave-one-out of the 66 eligible high-risk events, where persistence wins every time.

This freeze uses historical ESA-supported missions from 2015 to 2019, not live catalogues. The local test has nine events at $\log_{10} P_c \geq -6$ and three at -5; the official test has 150 at -6, which is why F_2 is estimable there. Manoeuvres are out of scope. SHAP explains the fitted model, not physical causation. Official-test MAE wins without the persistence guard do not improve ESA-style loss (H3). Changing the class cut without retuning does not create operational warning skill: the persistence–ensemble tie at -6 remains at -5, and the single -4 positive is missed. H4 is mixed: covariance volume tracks later movement; the max-risk gap tracks floor membership, not extra movement. That is a property of the CDM fields, not a covariance-tasking model [7].

Next measurements are publication figures with SI-style axes, and horizon intervals from the redraws.

ACKNOWLEDGMENT

This work uses the ESA Space Debris Office Collision Avoidance Challenge dataset [3], [4]. ESA, ISRO, NASA, ISTRAC, and CCSDS did not endorse this project.

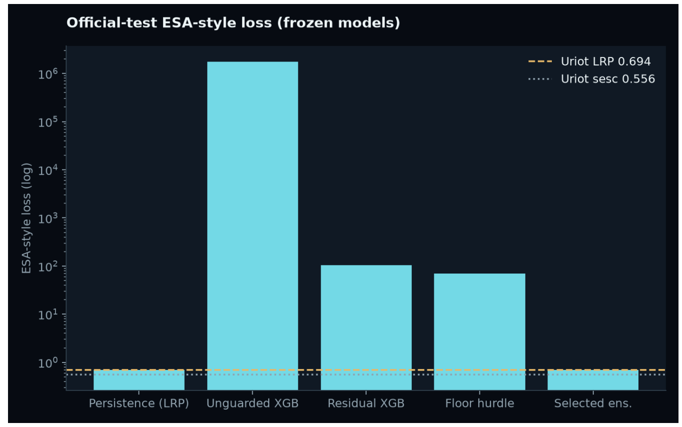

Fig. 3. Official-test ESA-style loss for frozen models versus Uriot last-risk-prediction (0.694) and sesc (0.556). Log scale.

TABLE V
FROZEN TEST BY TRAIN-QUARTILE OF
$\texttt{DILUTION\_GAP} = \texttt{MAX\_RISK\_ESTIMATE} - \texttt{RISK}$ ($n = 1659$).

Q	n	Floor	$\lvert\Delta\text{risk}\rvert$	XGB MAE
1	410	0.556	13.03	7.35
2	416	0.781	6.15	2.90
3	417	0.964	0.52	0.69
4	414	0.954	0.75	0.37

REFERENCES

[1] Consultative Committee for Space Data Systems, *Conjunction Data Message (CCSDS 508.0-B-1)*, 2013.
[2] M. D. Hejduk and L. C. Johnson, "Satellite conjunction assessment risk analysis for dilution region events: Issues and operational approaches," in *Space Traffic Management Conference*, 2016.
[3] T. Uriot, D. Izzo, L. F. Simões, R. Abay, N. Einecke, S. Rebhan, J. Martinez-Heras, F. Letizia, J. Siminski, and K. Merz, "Spacecraft collision avoidance challenge: Design and results of a machine learning competition," *Astrodynamics*, vol. 6, pp. 121–140, 2022, arXiv:2008.03069.
[4] ESA Advanced Concepts Team and Space Debris Office, "Collision avoidance challenge dataset," Zenodo, 2021.
[5] M. Duncan, J. Wysack, and J. Frisbee, "Collision probability forecasting using a Monte Carlo simulation," in *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*, 2014.
[6] S. Alfano, "Relating position uncertainty to maximum conjunction probability," in *AAS/AIAA Astrodynamics Specialist Conference*, 2003, AAS 03-548.
[7] J. Stauch, T. Olson, C. Reid, C. Grey, D. C. Kesler, and B. G. Marchand, "Contextual predictive model for early identification of high-covariance conjunctions," *The Journal of the Astronautical Sciences*, vol. 73, p. 19, 2026.
[8] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.

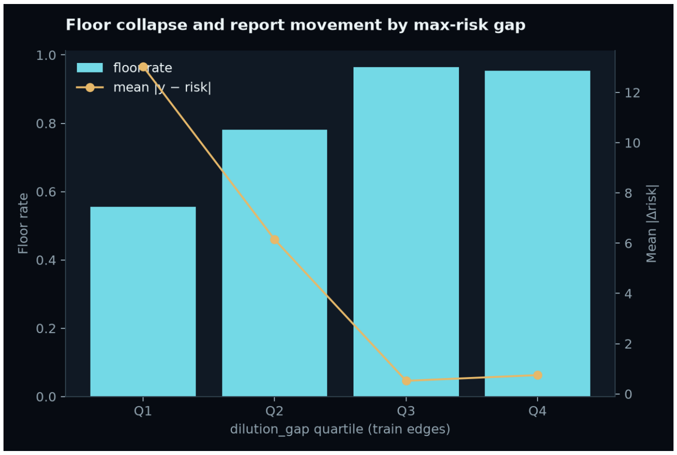

Fig. 4. Floor rate and mean $|y - \text{risk}|$ by train-quartile of the max-risk gap (frozen test).

TABLE VI
FROZEN LOCAL TEST ($n = 1659$). SAME PREDICTIONS; CLASS CUT t ONLY. P IS PERSISTENCE; E IS THE SELECTED ENSEMBLE. FR IS ACCEPTED $\widehat{y} < t$ WHILE $y \geq t$ UNDER THE EXISTING -6 ABSTENTION MASK. L OMITTED WHEN $F_2 = 0$.

t	$n+$	P F_2	P FR	P L	E F_2	E FR	E L
-8	68	0.660	2	1.66	0.210	5	9.49
-7	37	0.502	3	0.777	0.215	4	2.51
-6	9	0.361	1	0.167	0.361	1	0.167
-5	3	0.349	0	0.032	0.349	0	0.032
-4	1	0	1	—	0	1	—